

Engineering Mathematics: Rank and Nullity via Echelon Form

This study note covers the fundamental process of finding the **Rank** and **Nullity** of a matrix using **Row Echelon Form**. This is a critical skill for understanding the dimensionality of linear systems and is a prerequisite for advanced Engineering Math.

1. Core Concepts

What is Row Echelon Form?

A matrix is in Row Echelon Form (REF) if:

1. All non-zero rows are above any rows of all zeros.
2. The leading coefficient (the first non-zero number from the left, called a **pivot**) of a non-zero row is always to the right of the leading coefficient of the row above it.
3. All entries in a column below a leading coefficient are zeros.

Rank and Nullity

- **Rank (ρ):** The number of non-zero rows in the Row Echelon Form of a matrix. It represents the number of linearly independent rows.
- **Nullity (ν):** The number of free variables in the system.
- **Rank-Nullity Theorem:** For any matrix with n columns:

$$\text{Rank} + \text{Nullity} = n$$

2. Step-by-Step Algorithm

To transform a matrix into Echelon form, we use **Elementary Row Operations**:

1. Swap two rows ($R_i \leftrightarrow R_j$).
 2. Multiply a row by a non-zero constant (kR_i).
 3. Add or subtract a multiple of one row to another ($R_i \rightarrow R_i \pm kR_j$).
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3. Detailed Solutions

Problem (i)

$$A = \begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ 1 & 3 & 4 & 1 \end{bmatrix}$$

Step 1: Use R_1 to eliminate the first elements of R_2 and R_3 .

Perform $R_2 \rightarrow R_2 - 3R_1$ and $R_3 \rightarrow R_3 - R_1$.

$$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 3 & 9 & 12 & 9 \\ 1 & 3 & 4 & 1 \end{bmatrix} (R_2 \rightarrow R_2 - 3R_1) \longrightarrow \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 0 \\ 1 & 3 & 4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 0 \\ 1 & 3 & 4 & 1 \end{bmatrix} (R_3 \rightarrow R_3 - R_1) \longrightarrow \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix}$$

Step 2: Swap R_2 and R_3 to bring the non-zero row up.

$$\begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -2 \end{bmatrix} (R_2 \leftrightarrow R_3) \longrightarrow \begin{bmatrix} 1 & 3 & 4 & 3 \\ 0 & 0 & 0 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Final Result for (i):

- **Non-zero rows:** 2
 - **Rank (ρ):** 2
 - **Columns (n):** 4
 - **Nullity (ν):** $4 - 2 = 2$
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Problem (ii)

$$B = \begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$$

Step 1: Eliminate elements below the pivot in the first column.

Perform $R_2 \rightarrow R_2 - R_1$, $R_3 \rightarrow R_3 - 2R_1$, and $R_4 \rightarrow R_4 - 3R_1$.

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 1 & 3 & 2 & 2 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix} (R_2 \rightarrow R_2 - R_1) \longrightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 2 & 4 & 3 & 4 \\ 3 & 7 & 4 & 6 \end{bmatrix} (R_3 \rightarrow R_3 - 2R_1) \longrightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 3 & 7 & 4 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 3 & 7 & 4 & 6 \end{bmatrix} (R_4 \rightarrow R_4 - 3R_1) \longrightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

Step 2: Eliminate elements below the second pivot.

Perform $R_4 \rightarrow R_4 - R_2$.

$$\begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} (R_4 \rightarrow R_4 - R_2) \longrightarrow \begin{bmatrix} 1 & 2 & 1 & 2 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Final Result for (ii):

- **Non-zero rows:** 3
- **Rank (ρ):** 3
- **Columns (n):** 4
- **Nullity (ν):** $4 - 3 = 1$